

Parliamentary Elections, Political Connectedness and Firm Returns: Evidence from India

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Abstract

The present research paper investigates the political connectedness of the firms in India and evaluates its impact on their financial performance. The research study includes the analysis of select firms belonging to various sectors like Bank, FMCG, Infrastructure, Metal, Media and Pharmaceutical industry. To understand the political connect of the firms, four Lok Sabha election results are taken into consideration including 16th, 15th, 14th and 13th election results announcement. In this paper, CNX Nifty is taken as the benchmark market index and connected with the political affiliations. The research paper concludes that politically connected firms are often larger in size, carry less risk and outperform the unconnected firms especially in terms of stock market performance.

Key Words: *Parliamentary Elections, Political Connectedness, Firm Returns*

1. INTRODUCTION

Elections matter for an economy and have an impact on the way it functions. Partisan theory by Hibbs (1977) laid down the premise that different political parties choose different inflation/unemployment combinations systematically and thus changes in the partisan orientation of the party in power affect the economy. Later, the rational partisan theory (Alesina, 1987) extended the premise by stating that when partisan differences are expected rationally, then people's economic behaviour is adjusted prior to each election based on the changes in the probability of a party's win. More recent studies suggested that if political parties win by a wide margin then they are more likely to implement their favourite policies after the election (Conley 2001; Fowler 2005; Fowler and Smirnov 2005). In a democratic economy, where election outcomes may sway in any party's direction, formation of alliances becomes imperative for favourable policy implementation. However, if there is more uncertainty with respect to the margin of victory for any one party then policy uncertainty also increases. Also, the formation of coalitions after the elections aggravate this uncertainty. Due to this increase in uncertainty, the expectations of people might lead to underconfidence/overconfidence, which might in effect lead to fall/rise in credit ratings for the country, fall/rise in investor confidence, decline/increase in foreign investments etc. So, in a democratic economy, it becomes imperative for businesses to hedge these risks. Businesses would therefore, try and form relationships with political parties so that they could take advantage of the favourable policies once a party comes in power. These relationships can be built over a period of time and result in forming expectations about the businesses. Corporate political activities have been globally examined by scholars (Baysinger, Keim and Zeithaml, 1985; Getz, 1993; Keim and Baysinger, 1988; Pittman, 1977; Zardkoohi, 1985). In a transnational-study, Faccio (2006) documented political connections in over 74 percent of the countries in her sample.

As these connections are built over time, the benefits of political connections may come in many forms and materialise slowly.⁸ The benefits to corporates are however subject to measurement issues as these tend to be a lot more qualitative in nature – and of course, often invisible. To avoid these problems many researchers have followed the approach of looking at stock market reactions to announcements of political appointments. Stock-market reactions are immediate, which makes it easier to relate cause and effect and abstract from confounding factors. Further, stock market reactions are a comprehensive measure of potential benefits (Luechinger and Moser, 2012). Also, one needs to note that political connections are but one among a myriad of factors of firm performance. Fisman (2001) attempts to provide a measure of the value derived from such connections and tries to give us a possible index for political connectedness. While the value of political connection to firms has received considerable research interest (Ang and Boyer, 2000; Ang, Ding and Thong, 2010; Fisman, 2001; Goldman, Rocholl and So, 2009), the question of whether political connection enhances firm value has mixed findings. This study is an attempt to bring together these different fields of research and provide a possible link between stock markets and political connectedness. It accomplishes this by using a democratic setting of Indian economy.

India has parliamentary elections every five years and for the last four elections 1999, 2004, 2009 and 2014, the results have been swaying in favour of different parties. The outcome of three of these elections has led to formation of coalition governments and only one election resulted in a complete majority. Indian stock markets reacted accordingly showcasing the expectations of the investors. For instance, after the 2009 election results, for the first time in the history of Indian stock markets, trading was halted at the upper circuit one minute after trading began. Businesses in India therefore, lose or gain value on their stocks around the announcement of the election results. Investors might form expectations based on the relationship of businesses with the political parties and accordingly, rationalize their decisions of buying or selling a stock.

In recent years, Indian media has increased their efforts in terms of providing more information about political parties and their business affiliations too. There has been increased interaction between the two, and many of such relationships have been more open and public than others. It has been perceived that in India there are a few companies that enjoy significant relationships with certain political parties. Over the years such benefit-seeking relationships have taken different forms such as lobbying, reporting constituency building, and personal service. Historically, Indian businessmen have shown these associations by either openly supporting a political party or by being formally silent about their inclination. For instance, G D Birla, one of the well-known business man of India, openly identified himself with a major political party, the Congress party and it is believed that along with other prominent Marwari industrialists he became one of the major financial contributors of the party. From those times, of course, a lot has changed in politics as well as in business. India has witnessed a growth in the number of political parties and more so, coalitions. In terms of businesses, we have seen India emerging as a growth destination with more businesses coming to the forefront. With the advent of computers, and specially the internet driven social media, even the amount of information available on these political links of business firms has increased.

Simultaneously, Indian stock markets have also become more efficient. It can be presumed that these factors might have driven investors' sentiments around elections. This study tries to evaluate these expectations with two objectives. First, to understand whether the financial markets' factor in relationships in the specific stock price of companies with known relationship with political parties; and secondly, also to see, if it is possible to capture the assumptions made by capital market on some of the more unknown relationships. These are accomplished by studying anomalies in stock returns, after declaration of general election results.

While, in a democratic nation like India, members are free to have political affiliations according to personal beliefs, conflicts of interests arise when this closeness can be used to benefit certain

corporate entities. More importantly, it is possible that the market may believe that a particular political association maybe beneficial to a particular company. Hence, political events, associations, elections provide a setting for ascertaining firm's stock market performance and establishing a possible connect with the event in question.

Our contributions to the current literature are threefold. One, the motivation behind the study, is to understand if the Indian stock market factors the information related to election results, time required to form the coalition and the composition of the coalition. Two, this study breaks into a relatively new area, especially in the Indian context where we attempt to look beyond the impact on the market as a whole and dig deeper into the effect of perceived political-corporate alliances on firm's stock performance and three, given the three elections that we have identified, we feel that the coalition politics can be clearly identified due to the shift in the impact on the stock performance of the firm when there are different parties coming to power.

The rest of the paper is structured as follows. Section 2 presents the relevant literature. Section 3 describes our data and methodology. Section 4 presents the results, Section 5 provides the implications of the results and section 6 concludes.

2. LITERATURE REVIEW

Many researchers have looked at the stock market reactions to gauge this positive/negative impact of the political events and relationships. Several research studies have looked at an overall impact of elections news on the benchmark stock market index (Brander,1989; Forsythe, Nelson, Neumann and Wright, 1992; Vuchelen, 2003). Most consider US data of presidential elections. Foerster (1994) finds effects on Canadian stock returns of Canadian as well as of American government changes. The transnational effect of US elections on the stock market index of 18 countries is supported by

Foerster and Schmitz (1997). Pantzalis, Stangeland and Turtle (2000) found elections effects in the stock market returns of 33 countries. The point to be made here is that most of the time only the effects of the actual election are considered, and that the election results are not (the result of the incumbents or the ideologies of the election winners). Some exceptions exist, but the results are mixed. Our study fills this gap by choosing election results as the event and factoring in the time to form coalitions in our event window.

Lobo (2000) examines the impact of U.S. elections and partisan politics on the stock market using jump–diffusion models of daily stock returns from 1965 to 1996. Leblang and Mukherjee (2005) construct a model of speculative trading to examine how the mean and volatility of stock prices is affected both by government partisanship and by traders' expectations of electoral victory by the right-wing or left-wing party. A number of studies have been undertaken in the US to understand political influences on stock markets including Beyer and Johnson (2004), Booth and Booth (2003) , Nofsinger (2007), Johnson, Chittenden and Jensen (1999), Powell, Smith, and Whaley (2007) and Santa Clara and Valkanov (2003). However, almost all of these studies look at the relation between business and politics as a single unit and thus give us a generic scenario, while, in reality, the impact of such events can be different for different firms on the basis of the relationships which they develop with the parties. Also, there might be differences on the basis of the distribution of ministries in the coalition. Businesses might show a positive/negative value depending on their association with the party which has been assigned the ministry which has a direct impact on the functioning of the business/industry.

There have been some studies that have considered industry wise analysis of the event. Knight (2006) analyzes the stock return around the 2000 US Presidential Election of a sample of firms to show that policy platforms are capitalized into equity prices. He also uses the firms' donation data as

a robustness check for the categorization made by the financial analysts, the thrust of his work is that firms in industries, which stand to benefit from the Bush administration, will donate more to the Bush Campaign. Similarly, Shon (2004) looks only at donations at an industry level and shows that firms in industries that donate more to Republicans exhibit a positive stock price return following Bush's win in the 2000 election. Jayachandran (2006) looks at the 2001 departure of Senator Jim Jeffords from the Republican to the Democrat party and shows that this event results in a decrease in value of firms donating to the Republicans.

These studies highlight the importance of looking at firm level data to understand the impact of the political event. For our research question, we undertake a similar analysis for India where we present the results firm wise for different industries so that we capture the impact on the firm depicting the kind of relationships the market perceives the firm has with the coalition parties in power.

Our research question looks at India because of several reasons. From the governance perspective, India follows a parliament based system unlike the US. India is also a nation with one of the largest political democracy. While a government can be formed by a party with majority, the recent governments have taken to coalitions to reach the requisite majority. Therefore, it becomes imperative to factor in the dynamics of several parties coming to power. Vuchelen (2003) in his paper tries to differentiate the impact of political events on the stock market between a majority based electoral system (single party governments) and proportional representation (coalition government). He mentions in his paper that important political events occur more frequently in coalition systems as compared to two-party systems because the election outcome and the disappearance of uncertainty about future policies no longer coincide. In coalition systems, four political events can affect the stock market – the election results, time required to form the coalition, composition of the coalition and the new government's policies. Investors form expectations about each one.

In a coalition, it also matters as to which party is handling which ministry/industry and to that extent, political connectedness becomes uncertain. We think that India provides us a good setting in terms of timeline and in terms of different coalitions coming to power over a long period. Thus, we get a chance to look at positive negative impact shifts when the political results shift from one coalition to another.

3. DATA AND METHODOLOGY

The focus of this study is to establish a possible link between a democratic country's most significant political event and stock returns of the corporates in order to work towards a metric which helps in understanding the impact of such an event on the firms operating therein. Firstly, we identify the event and the estimation windows for our study, then we test for the impact of the events on the stock returns of the firms.

For the first part of our study, we chose three events : 13th, 14th, 15th and 16th Lok Sabha elections. We provide a summary of these events in Table I.

Table I. Events and dates used in the study

Election	Date of the election results	Lead Party	Other Key Coalition Members of the Ruling Party
13 th Lok Sabha Election	6 October 1999	Bhartiya Janata Party (BJP)	National Democratic Alliance+Telugu Desam Party*
14 th Lok Sabha Election	13 May 2004	Indian National Congress (INC)	United Progressive Alliance**
15 th Lok Sabha Elections	18 May 2009	Indian National Congress (INC)	United Progressive Alliance***
16 th Lok Sabha Elections	16 May 2014	Bhartiya Janata Party (BJP)	National Democratic Alliance

*The member parties of the alliance included Bharatiya Janata Party, Dravida Munnetra Kazhagam, Telugu Desam Party--external support, Samata Party, Biju Janata Dal, Shiromani Akali Dal, All India Trinamool Congress, Shiv Sena, Pattali Makkal Katchi, Lok Shakti, Marumalarchi Dravida Munnetra Kazhagam, Haryana Vikas Party, Janata Party, Mizo National Front, Sikkim Democratic Front.

** The member parties included Indian National Congress, Rashtriya Janata Dal, Dravida Munnetra Kazhagam, Nationalist Congress Party, Pattali Makkal Katchi ,Telangana Rashtra Samithi , Jharkhand Mukti Morcha, Marumalarchi Dravida Munnetra Kazhagam, Lok Jan Shakti Party, Jammu & Kashmir Peoples Democratic Party, Republican Party of India, Muslim League Kerala State Committee, Republican Party of India (Athvale) , Arunachal Congress, Kerala Congress (Mani), Indian Union Muslim League.

***The member parties included Dravida Munnetra Kazhagam, Nationalist Congress Party, Rashtriya Lok Dal, Jammu & Kashmir National Conference, Jharkhand Mukti Morcha, Indian Union Muslim League, All India Majlis-e-Ittehadul Muslimeen, Kerala Congress (Mani), Viduthalai Chiruthaigal Katchi. The Supporting parties for Lok-Sabha 2009 included Samajwadi Party, Bahujan Samaj Party, Rashtriya Janata Dal, Janata Dal (Secular), Nagaland People's Front, Bodoland People's Front, Swabhimani Paksha, Bahujan Vikas Aaghadi, All India United Democratic Front, Sikkim Democratic Front.

****The member parties included Shiv Sena, Swabhimani Paksha, Republican Party of India (Athavale), Rashtriya Samaj Paksha, Desiya Murpokku Dravida Kazhagam, Pattali Makkal Katchi, Marumalarchi Dravida Munnetra Kazhagam, Kongunadu Makkal Desia Katchi, Indhiya Jananayaga Katch Kerala Congress (Nationalist), All India N R Congress, Telugu Desam Party, Lok Janshakti Party, Rashtriya Lok Samata Party, Revolutionary Socialist Party (Bolshevik), Shiromani Akali Dal, Haryana Janhit Congress, Apna Dal, National People's Party (India), Naga People's Front, United Democratic Front, Manipur People's Party, Gorkha Janmukti Morcha, Maharashtrawadi Gomantak Party, Jana Sena Party, New Justice Party.

To understand the impact of our political event, the firms that we use in our sample belong to different sectors namely infrastructure, banking, fast moving consumer goods (FMCG), pharmaceutical, metal and media. As one can observe, the selection of these industries has been done to capture the width in both, the regulation and the relative dominance of the public sector vis-à-vis the private sector.¹ The firms are the constituents of the sectoral indices of National Stock Exchange (NSE). The list of all the firms is provided in the appendix. A total of 89 firms listed on NSE were included in the analysis. The data is collected from Centre for Monitoring Indian Economy (CMIE) Prowess database. We calculate daily returns for the firms using the stock prices. CNX Nifty is taken as the benchmark market index for the same period.

¹ Gowda and Sridharan (2011) while interviewing with industry chief executive officers (CEOs) over the past few years (both before and after the 2009 national election), find that government permissions/clearances/licensing powers became pressure points for extorting payments from businesses by political parties. They find it evident especially in the mining and real estate development industries and also in large industrial projects. They add that, 'Industry, being

vulnerable to such discretionary regulation and extortion, is averse to alienating parties that are in power or that may come to power.’

Event study methodology has been used in most of the studies assessing the impact of an event on the stock returns of the companies. To show the impact of election results on the sample of firms we first calculated Abnormal Returns using the market model. Cumulative Abnormal Returns are then calculated for each company within the sector (Mackinlay, 1997). Following models were used to calculate the same:

Equation I: Abnormal Returns

$$AR_{i\tau} = R_{i\tau} - \hat{\alpha}_i - \hat{\beta}_i R_{m\tau} \dots \dots \dots (I)$$

where

AR_i = Abnormal Returns of the firm i in the event window for day τ ($i = 1, 2, 3 \dots N$)

τ = Returns are indexed in event time using τ ($\tau = T1 + 1, \dots, T2$)

$R_{i\tau}, R_{m\tau}$ = Return in event period τ for security i and market respectively

α_i, β_i = Parameters of the market m

Equation II: Average Abnormal Returns

$$AAR_i = \frac{1}{N} \sum_{i=1}^N AR_{i,\tau} \dots \dots \dots (II)$$

Where

$AR_{i,\tau}$ = Abnormal Returns of the firm i in the event window ($i = 1, 2, 3 \dots N$)

Equation III: Cumulative Abnormal Return

$$CARI(\tau_1, \tau_2) = \sum_{\tau=\tau_1}^{\tau_2} AR_{i\tau} \dots \dots \dots (III)$$

where

$CARI(\tau_1, \tau_2)$ = cumulative abnormal return from τ_1 to τ_2 where $T1 \leq \tau_1 \leq \tau_2 \leq T2$

$AR_{i\tau}$ = Abnormal Return of the firm i in the event window

However, our event, election results, is an event on the same day for all the firms and hence, provides a case of clustering. To handle clustering, Mackinlay (1997) discusses two approaches. First is to aggregate abnormal returns into a portfolio dated using event time and then perform the event study on the portfolio. The second method is to test the null hypotheses of the event having no impact using unaggregated security by security data. Since we are dealing with total clustering, the second method becomes the preferable methodology. We use an application of multivariate regression model with dummy variables for the event date (Schipper and Thompson, 1983, 1985, Collins and Dent, 1984).

Schipper and Thompson's (1983) approach explicitly incorporates heteroskedasticity in the estimation process. It also makes more efficient use of the available data and is particularly suitable for studies involving relatively small sample sizes. Another clear advantage of this approach over the standard event study methodology is that it allows testing of joint hypotheses as well as tests about average and cumulative excess returns (Saunders and Angbazo, 1996). In this approach, daily firm returns are estimated using an expanded version of the market model with a zero-one dummy variable appended to reflect the election results announcement over the three events.

$$\tilde{R}_{it} = \beta_{i0} + \beta_{im} \tilde{R}_{mt} + \sum_{k=1}^K \beta_{ik} D_{kt} + \tilde{\epsilon}_{it} \dots\dots\dots(IV)$$

where

$\tilde{R}_{it}$ = rate of return on firm i for day t (i = 1,2,3....N)

β_{i0} = event independent constant term

β_{im} = systematic risk or sensitivity of the firm's rate of return to changes in market's rate of return

$\tilde{R}_{mt}$ = S & P CNX Nifty rate of return (market rate of return) for day t

β_{ik} = sensitivity of firm i's rate of return to event k

D_{kt} = dummy variable reflecting the occurrence or non - occurrence of the event, takes the value of 1 on the event day and zero otherwise

$\tilde{\epsilon}_{it}$ = random disturbance term which is assumed to be independent of the market return, serially independent and normally distributed

We create dummy variables for two event windows (-1,+3) and (-1,+5) to capture the impact of election results and the sentiment with respect to political coalitions thereafter⁹. Our hypotheses tests whether an event had a significant impact on each firm in the portfolio individually, i.e., is each firm's event parameter β_{ik} equal to zero.

$$H_0 : \beta_{1k} = \beta_{2k} = \beta_{3k} = \dots\dots\dots = \beta_{Nk} = 0 \dots\dots\dots(II)$$

This test is more likely to reject the null hypothesis of zero impact if an event influences only a few firms or influences some firms positively and others negatively. For the purpose of creating a political index, we identify the significance of the event dummy for each of the firms and classify them into categories on the basis of the strength of the significance. If the dummy coefficient is significant at 1percent level then we name it as having 'high' degree of relationship, at 5percent we call it 'medium' and at 10percent we call it 'low'.

4. RESULTS: In this section, the result of our studies will be presented in two sub-sections.

(A) Calculation of Abnormal Returns, Average Abnormal Returns and Cumulative Abnormal Returns both election wise and sector wise.

Figure 1: Average Abnormal Return (Election Wise)

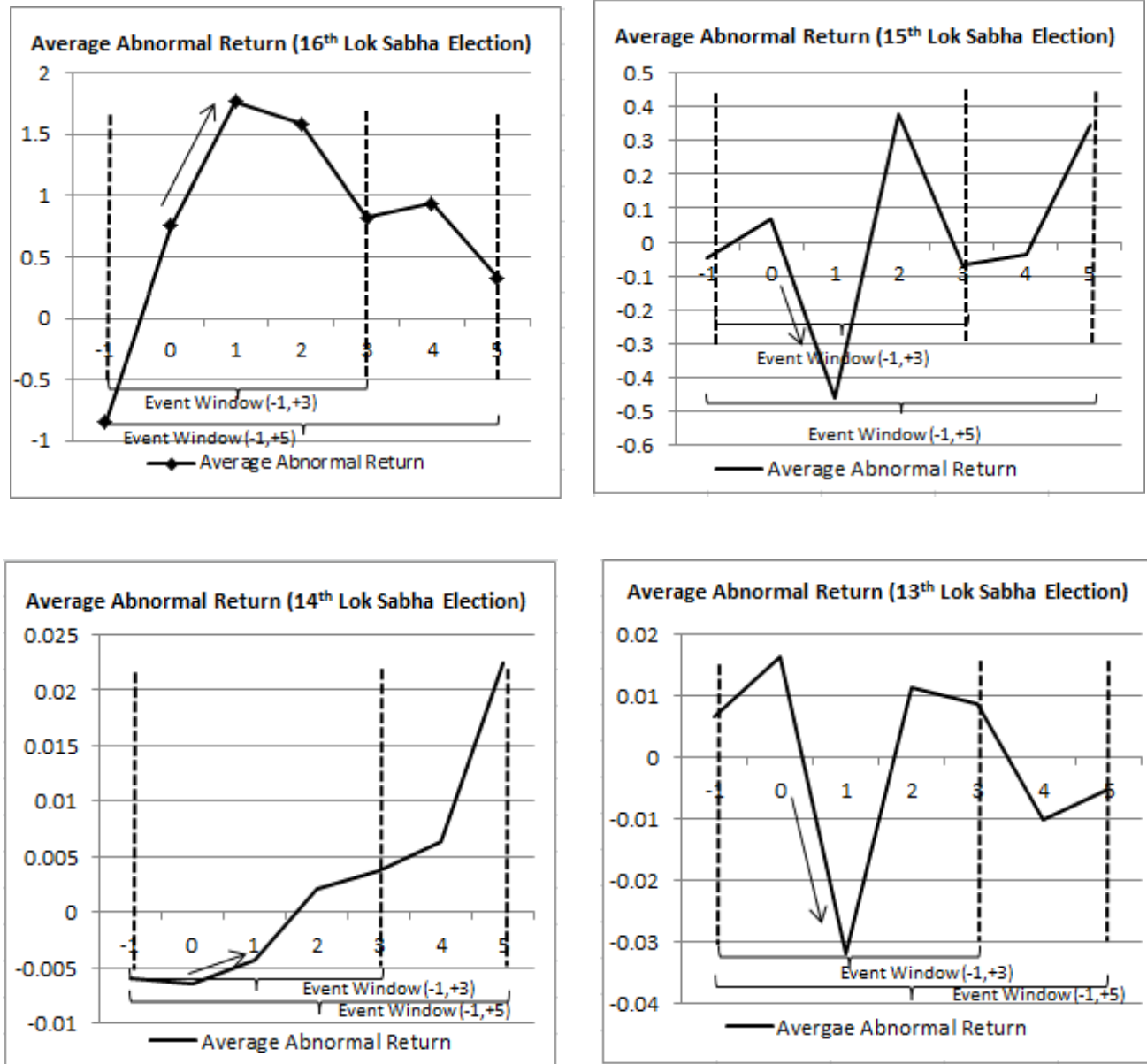


Figure 1 provides the consolidated Average Abnormal Returns for both first event window (-1,+3) and second (-1,+5) for all the firms election wise.

Figure 1 presents the average abnormal returns of all the firms in 6 sectors election wise. The results present a consolidated average abnormal return for both event window 1 (-1,+3) as well event window 2 (-1,+5). During the 16th Lok Sabha election, the announcement of the election results show a positive impact on the average abnormal return, wherein BJP came into power with majority. However, days after the announcement the average abnormal return shows a slight downfall. The average abnormal return during the 15th Lok Sabha election depicts a negative reaction post the announcement of the election results. However, the AARs start gaining back and grow positively after few days of the announcement. The 14th Lok Sabha election wherein the UPA formed a coalition government shows a stable positive growth in the average abnormal return. However, during the 13th Lok Sabha election the average abnormal return shows a sharp decline immediately after the results announcement. However, the AARs start showing positive results after some time. The possible reason for this could be the formation of coalition party or the reformation of policies, which may have resulted in gaining back the trust of the investors.

Figure 2: Average Abnormal Return (Sector Wise)

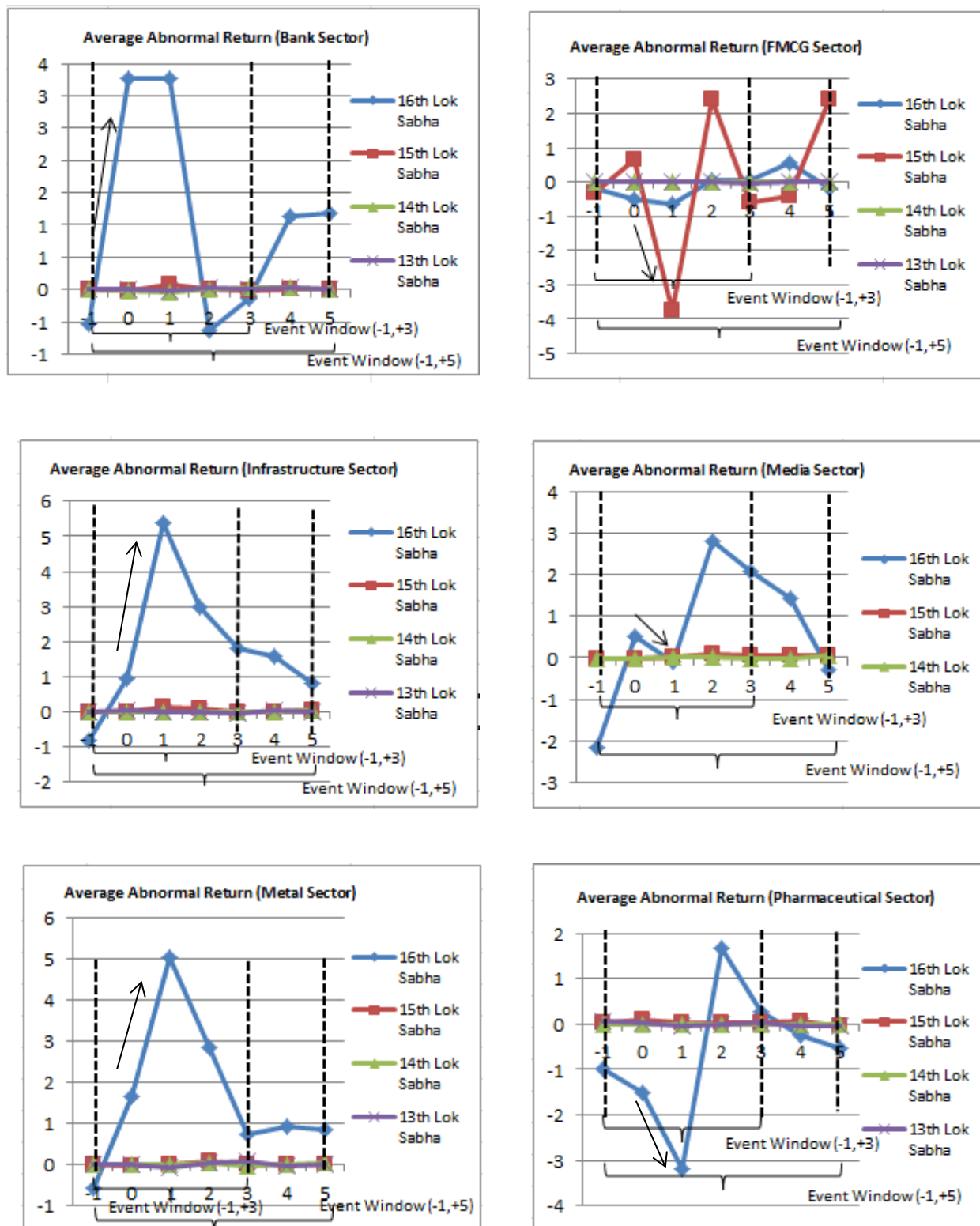


Figure 2 provides the consolidated Average Abnormal Returns for all the firm in the sample for all the four elections sector wise for both the first event window (-1,+3) and second event window (-1, +5 days).

Figure 2 presents the consolidated average abnormal returns for all the firms across 6 sectors and 4 election results sector wise. The results show a consolidated average abnormal return for the event window (-1, +3) i.e. first dummy variable and (-1,+5) second dummy variable. The average abnormal returns for the Bank Sector remains positive for some time after the announcement of the 16th Lok Sabha election results. However, it shows a slight downfall post announcement period, which again starts showing positive results after few days. During the 15th, 14th and 13th Lok Sabha election results, the return remains very low and almost near to 0.

A similar trend is seen for infrastructure and metal sector wherein the average abnormal return shows a positive growth immediately post the announcement day (i.e. day 0). Media reports suggest that the possible reason could be the assurance of the BJP Govt for improvised infrastructure facility, better policy reformation and lucrative investment opportunities if the party comes in power (The Economist, 2015). The same may have created an optimistic belief for the investors to see an improved health of both the sectors and hence the AARs show a very positive reaction immediately after the election results announcement. The AARs do not show any significant results for the 15th, 14th and 13th Lok Sabha elections.

The average abnormal returns for the FMCG sector remains mostly low in the 16th, 14th and 13th Lok Sabha elections. However, in the 15th Lok Sabha elections, the average abnormal return does not remain stable and experiences points of extreme lows and highs. The plausible reason could be the covert relations of many FMGC giants with the ruling parties. For example: ITC, the leading tobacco production company which shows positive results with high significance made huge donations to all the major political parties (Perappadan, 2012).

For the Media and Pharmaceutical sector, the average abnormal return drifts down slightly after the announcement of the 16th Lok Sabha election results. However, it starts showing an upward trend after few days of the result. Especially the Pharmaceutical sector shows a significant positive result

post few days of the election results announcement. This might be due to the declaration of the special proposal plan of the BJP government to invest in the Pharmaceutical sector, modernize hospitals, upgrade infrastructure and adopt world class technology (BJP Manifesto, 2014) creating a positive wave for the investors to invest in this sector. The AARs of the 15th, 14th and 13th Lok Sabha elections do not show any significant results.

Table II: Cumulative Abnormal Return

Company Name	16th Lok Sabha Elections	15th Lok Sabha Elections	14th Lok Sabha Elections	13th Lok Sabha Elections
Bank Sector				
Axis Bank Ltd.	7.6953***	-0.0328	0.0690	0.003
Bank Of Baroda	-5.499	0.1258***	0.1005	0.1811***
Bank Of India	18.2504***	0.1555***	-0.0306	0.0108
Canara Bank	29.4890***	0.0634***	0.0097	
Federal Bank Ltd.	14.7666***			
H D F C Bank Ltd.	-2.4273	-0.0144	0.0770	-0.0246
I C I C I Bank Ltd.	-1.5669	0.0129	0.00000	0.0651
Indusind Bank Ltd.	-1.0668			
Kotak Mahindra Bank Ltd.	-2.2649			
Punjab National Bank	17.0542***	-0.0305	-0.0101	
State Bank Of India	15.4694***	0.1247***	-0.0563***	0.0878***
Yes Bank Ltd.	0.6644	0.0904		

Company Name	16th Lok Sabha Elections	15th Lok Sabha Elections	14th Lok Sabha Elections	13th Lok Sabha Elections
FMCG Sector				
Britannia Industries Ltd.	-0.0521***		-0.0166	-0.1285***
Colgate Palmolive (India) Ltd.	-4.634***	-4.3786***	-0.0286	-0.0771***
Dabur India Ltd.	-5.3978***	-0.8547***	0.0095	-0.0988***
Emami Ltd.	0.0243			
GlaxoSmithkline Consumer Healthcare Ltd.	0.027		0.0491***	-0.0609***
Godrej Consumer Products Ltd.	-2.3137	7.1741***	0.0665***	
Godrej Industries Ltd.	0.0381			
Hindustan Unilever Ltd.	-5.2515***	-3.9359***	0.0575	-0.0579***

I T C Ltd.	-0.0893***	-6.9681***	0.0132	-0.0519***
Jubilant Foodworks Ltd.	0.0730***			
Marico Ltd.	1.4822		0.0484***	
Mcleod Russel India Ltd.		1.1921***		
Procter & Gamble Hygiene & Health Care Ltd.	5.1798			
Tata Global Beverages Ltd.	0.0407***	-2.4727***		-0.0196
United Spirits Ltd		15.3162***	-0.023	

Company Name	16th Lok Sabha Elections	15th Lok Sabha Elections	14th Lok Sabha Elections	13th Lok Sabha Elections
Infrastructure Sector				
Ansal Properties & Infrastructure Ltd.	0.2607***	0.3555***		0.0995
D L F Ltd.	0.2616***	0.1165		
Era Infra Engg. Ltd.	0.2976***	0.2682***		
G V K Power & Infrastructure Ltd.	0.2477***	0.1942***		
GMR Infrastructure Ltd	0.1768***	0.2014***		
Gammon India Ltd.	41.4043***	0.5126***	-0.0941***	-0.0162
Hindustan Construction Co. Ltd.	33.1324***	0.3656***	0.0233	-0.1030***
Housing Development & Infrastructure Ltd.	0.1252***	0.2197***		
I V R C L Infrastructures & Projects Ltd	42.9668***	0.3160***	-0.0064	
Jaiprakash Associates Ltd.	0.1388***	0.0121		
Lanco Infratech Ltd	57.7924***	0.1814***		
Nagarjuna Constructions Ltd		0.2397***	0.0280	0.0008
Omaxe Ltd.	0.1016***	0.4099***		
Peninsula Land Ltd.	0.2609***	0.1419***	-0.0813***	0.0169
Unitech Ltd	0.4094***	0.1966***	0.0513	0.0721***

Company Name	16th Lok Sabha Elections	15th Lok Sabha Elections	14th Lok Sabha Elections	13th Lok Sabha Elections
Metal Sector				
Bhushan Steel Ltd.	-5.8837***	-0.0285		
Gujaratmineraldevpcorpnltd		0.0256	-0.0914***	0.1796
Gujaratnrecokeltd		0.1893***		
Hindalco Industries Ltd.	2.7534***	-0.0913	-0.0101	-0.0423
J S W Steel Ltd.	3.0228	0.0194		0.0186
Jindal Saw Ltd.	34.8857***			

Jindal Steel & Power Ltd.	11.0371***		0.1320***	
Maharashtra Seamless Ltd.	13.1021***			
monnetispatenergyltd		0.2110***		
National Aluminium Co. Ltd.	29.2863***	0.1317		
sesagoaltd		0.0039	-0.0227	-0.0056
steelauthorityofindialtd		0.0963***	0.0846***	0.1291
sterliteindustriesindialtd				-0.0491
Tata Sponge Iron Ltd.	-0.3558			
Tata Steel Ltd.	0.4111	0.1077	0.1565***	
var1		0.0598		
Vedanta Ltd.	26.6437***			
Welspuncorpltd		0.1666***		

Company Name	16th Lok Sabha Elections	15th Lok Sabha Elections	14th Lok Sabha Elections	13th Lok Sabha Elections
Media Sector				
Adlabs Films ltd		0.1321	-0.0117	
Balaji Telefilms ltd		0.3005***	0.0517	
Deccan Chronicle holdings ltd		0.2423***		
Den Networks Ltd.	4.0085			
Dish TV India ltd		0.1477***		
Eros International Media Ltd.	14.6983***			
Hathway Cable & Datacom Ltd.	18.1767***			
HT Media ltd		0.5138***		
Inox Leisure Ltd.	11.4859***			
Jagran Prakashan Ltd	0.4549	0.3037***		
Network Returns	-24.1616**			
New Delhi Television Ltd		0.2662***		
P V R Ltd.	6.4156			
Pyramidsaimiratheatreltd		0.1552***		
Sun T V Network Ltd.	11.1258***	0.1432***		
T V Today Network Ltd.	-7.2886			
Tv18 Broadcast Ltd.	15.2682***			
UTV Software Communications Ltd		0.1183***		
Zee Entertainment Enterprises Ltd.	-2.9988	0.1202***	0.0179	
Zee news ltd		0.2471		

Company Name	16th Lok Sabha Elections	15th Lok Sabha Elections	14th Lok Sabha Elections	13th Lok Sabha Elections
Pharmaceutical Sector				
Aurobindo Pharma Ltd.	-5.3461***			
Biocon ltd		0.2198		
Cadila Healthcare Ltd.	-4.9364			
Cipla Ltd.	-3.0002***	0.0634		-0.0913
Divi'S Laboratories Ltd.	-6.5312***	0.2295		
Dr. Reddy'S Laboratories Ltd.	-10.1827***	0.1701***	0.0989	-0.2312
Glaxosmithkline Pharmaceuticals Ltd.	2.6623			-0.0839
Glenmark Pharmaceuticals Ltd.	-8.8524**	0.4457	0.0955	
Lupin Ltd.	-6.2344***	0.1241		0.0624
Piramal Enterprises Ltd.	4.4934***	0.0894	-0.0442***	0.2121
Ranbaxy Laboratories ltd		0.3852*	0.0438	-0.1299
Sun Pharmaceutical Inds. Ltd.	-7.1513***	0.2032		
var1		-0.9744	-0.0943	

Figure 3 provides the consolidated Cumulative Abnormal Returns for all the firm in the sample for all the sectors and across four elections for the event window (-1, +3 days) and event window (-1,+5).

Figure 3 presents the cumulative abnormal return for all the firms sector wise across all 4 elections. The results present the consolidated figures of the cumulative abnormal return including both the first as well as second event window i.e. including (-1,+3) and (-1,+5) respectively. The cumulative abnormal returns for the Bank sector show positive results for most of the banks during the 16th Lok Sabha election. Banks like Axis Bank, Canara Bank, Punjab National Bank and State Bank of India show positive results with high significance. However, during the 15th , 14th and 13th elections the cumulative abnormal return is significant only for few companies.

The CARs of the FMCG sector shows positive results for many companies including Britannia Industries Ltd, Colgate Palmolive and Dabur India Ltd. during the 16th Lok Sabha elections. During the 15th Lok Sabha elections, most of the companies continue to show positive results with high significance, however there are very few companies which show significant results during the 14th and 13th elections.

The cumulative abnormal return for the Infrastructure sector shows positive results for most of the companies during the 16th Lok Sabha elections, which indicate an optimistic belief of the investors, post the win of Modi government. Especially companies like Ansal Properties and Infrastructure Ltd, Hindustan Construction Company Ltd., Gammon India Ltd etc shows positive results with high significance.

Similar results are also seen during the 15th Lok Sabha elections wherein companies like Ansal Properties and Infrastructure Ltd, Gammon India Ltd, Hindustan Construction Company Ltd show results with high significance. Secondary sources state that many of these companies share close allies with the political parties and also made substantial donation to the parties in power (Karunakaran, 2012). Example: GVK group founder Mr. GVK Reddy's son married the daughter of Congress MP T Subbarami Reddy. The owner of the Lanco group Mr. L Rajagopal is a Congress MP and is also the son-in-law of former Union minister P Upendra.

During the 14th Lok Sabha elections, very few companies show positive results with significance. Gammon India Ltd which showed positive significant results in 16th, 15th elections continued to show significant results but with negative result (Sood et al, 2011). After the 13th Lok Sabha election result announcement very few companies witnessed significant results.

The CARs of the media companies during the 16th Lok Sabha elections shows positive results for few companies like Eros International Media Ltd., Hathway Cable & Datacom Ltd. During the 15th Lok Sabha election results announcement, media giants like HT Media Ltd, New Delhi Television Ltd (NDTV) show positive result with high significance.

Reports from secondary source state that many of the media companies share close allies with the political parties in power or are covertly related to them. Example: Jagran Prakashan Ltd is headed by Mr. Mahendra Mohan Gupta, the former Member of Parliament (Samajwadi party) also his brother has always been in good affairs with the UPA led Congress party. Further, the company,

which is managed by Ms. Shobhana Bhartiya is the daughter of industrialist KK Birla and has been a Rajya Sabha member of the Congress Party till 2012 (Paul, 2008). NDTV's ex co chairperson Mr. Pranay Roy's wife Ms. Radhika Roy is the sister of Ms. Brinda Karat Rajya Sabha, MP (NewsLaundry, n.d.).

The pharmaceutical sector during the 16th Lok Sabha election showed significant positive results for companies like Aurobindo Pharma Ltd., Cipla Ltd, Divi's Laboratories Ltd. For the remaining election result announcement, very few companies reacted positively to the news. Ranbaxy Laboratories Ltd. showed positive result during the 15th Lok Sabha elections but with low significance. The 14th and 13th election result shows significant CARs only for few companies.

Table III: Multivariate Regression Results

PANEL A: Media Sector Firms Results								
Firm Names	13th Lok Sabha Elections	13th Lok Sabha Elections	14th Lok Sabha Elections	14th Lok Sabha Elections	15th Lok Sabha Elections	15th Lok Sabha Elections	16th Lok Sabha Elections	16th Lok Sabha Elections
	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2
Balaji Telefilms Ltd.	NA	NA	Not Significant	Not Significant	Not Significant	High	NA	NA
	NA	NA	-0.0050 (0.0139)	0.00769 (0.0117)	0.0266 (0.0202)	0.0450*** (0.0169)	NA	NA
Hathway Cable Datacom Ltd	NA	NA	NA	NA	NA	NA	Medium	High
	NA	NA	NA	NA	NA	NA	2.375**	2.671***

							(1.046)	(0.882)
H T Media Ltd.	NA	NA	NA	NA	High	High	NA	NA
	NA	NA	NA	NA	0.0703*** (0.0185)	0.0769*** (0.0153)		
Jagran Prakashan Ltd.	NA	NA	NA	NA	High	High	Not Sig nifi can t	Not Significa nt
	NA	NA	NA	NA	0.0436*** (0.0135)	0.0454*** (0.0113)	1.046 (2.513)	0.0668 (2.134)
New Delhi Television Ltd.	NA	NA	NA	NA	Not Significant	High	NA	NA
	NA	NA	NA	NA	0.00508 (0.0175)	0.0398*** (0.0146)	NA	NA

PANEL B: Banking Sector Firms Results								
Firm Name	13th Lok Sabha Elections	13th Lok Sabha Elections	14th Lok Sabha Elections	14th Lok Sabha Elections	15th Lok Sabha Elections	15th Lok Sabha Elections	16th Lok Sabha Elections	16th Lok Sabha Elections
	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2
Bank Of Baroda	Low	Medium	Not Significant	Not Significant	Low	Low	Not Significant	Not Significant
	0.0240* (0.0126)	0.0268** (0.0106)	-0.0134 (0.0153)	0.0149 (0.0128)	0.0252* (0.0132)	0.0188* (0.0112)	-1.075 (1.002)	-0.808 (0.851)
Bank of India	Not Significant	Not Significant	Not Significant	Not Significant	High	Medium	Not Significant	Medium
	0.0008 (0.0113)	0.0016 (0.00957)	-0.0172 (0.0127)	-0.0045 (0.0107)	0.0354*** (0.0135)	0.0233** (0.0115)	1.865 (1.253)	2.681** (1.056)
Canara bank	NA	NA	Not Significant	Not Significant	Not Significant	Not Significant	High	High
	NA	NA	-0.0199 (0.0148)	0.0014 (0.0125)	0.0150 (0.0125)	0.0095 (0.0106)	3.475*** (1.138)	4.333*** (0.948)

Federal Bank Ltd	NA	NA	NA	NA	NA	NA	Medium	Medium
	NA	NA	NA	NA	NA	NA	2.241** (0.996)	2.170** (0.843)
Punjab National Bank	NA	NA	Not Significant	Not Significant	Not Significant	Not Significant	High	High
	NA	NA	-0.0075 (0.0169)	-0.0015 (0.0142)	0.0035 (0.0113)	-0.0045 (0.00954)	2.9310*** (0.899)	2.5060*** (0.763)
State Bank Of India	Not Significant	Not Significant	Medium	Not Significant	Medium	Medium	Not Significant	High
	0.0169 (0.0104)	0.0130 (0.00882)	-0.0155** (0.00664)	-0.0084 (0.00561)	0.0267** (0.0103)	0.0187** (0.00875)	1.135 (0.744)	2.2730*** (0.620)
YES Bank	NA	NA	NA	NA	Not Significant	Not Significant	Not Significant	Not Significant
	NA	NA	NA	NA	0.0154 (0.0154)	0.0135 (0.0130)	-0.2670 (1.245)	0.0976 (1.057)

PANEL C: FMCG Sector Firms Results								
Firm Name	13th Lok Sabha Elections	13th Lok Sabha Elections	14th Lok Sabha Elections	14th Lok Sabha Elections	15th Lok Sabha Elections	15th Lok Sabha Elections	16th Lok Sabha Elections	16th Lok Sabha Elections
	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2
ITC	Not Significant	Not Significant	Not Significant	Not Significant	Medium	Not Significant	High	High
	-0.0067 (0.00816)	-0.0077 (0.00691)	0.0108 (0.00687)	0.0019 (0.00581)	-2.2740** (0.914)	-1.0600 (0.778)	-0.0165*** (0.00583)	-0.0131*** (0.00496)
Britannia Industries Ltd.	Not Significant	Medium	Not Significant	Not Significant	NA	NA	Low	Not Significant
	-0.0177 (0.0113)	-0.0190** (0.00956)	-0.00289 (0.00679)	-0.00248 (0.00571)	NA	NA	-0.0157* (0.00863)	-0.0076 (0.00736)

PANEL D: Metal Sector Firms Results								
Firm Name	13th Lok Sabha Elections	13th Lok Sabha Elections	14th Lok Sabha Elections	14th Lok Sabha Elections	15th Lok Sabha Elections	15th Lok Sabha Elections	16th Lok Sabha Elections	16th Lok Sabha Elections
	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2
Bhushan Steel ltd	NA	NA	NA	NA	Not Significant	Not Significant	High	High
	NA	NA	NA	NA	-0.0054 (0.0186)	-0.0043 (0.0158)	-1.2670*** (0.348)	-0.8650*** (0.298)
Jindal Saw Ltd	NA	NA	NA	NA	NA	NA	High	High
	NA	NA	NA	NA	NA	NA	6.0130*** (1.303)	5.1260*** (1.106)
National Aluminum Co. Ltd.	NA	NA	NA	NA	Medium	Not Significant	High	High
	NA	NA	NA	NA	0.0385** (0.0170)	0.0197 (0.0145)	3.1780*** (1.041)	4.3030*** (0.861)

Tata Steel	NA	NA	High	High	Not Significant	Not Significant	Not Significant	Not Significant
	NA	NA	0.0217*** (0.00775)	0.0233*** (0.00646)	0.0224 (0.0148)	0.0161 (0.0126)	-0.146 (0.974)	0.0604 (0.827)
Vedanta Ltd	NA	NA	NA	NA	NA	NA	High	High
	NA	NA	NA	NA	NA	NA	4.3360*** (1.181)	3.9150*** (0.999)
Gujarat Mineral Development Corporation Ltd.	Not Significant	High	Medium	Not Significant	Not Significant	Not Significant	NA	NA
	0.02790 (0.0191)	0.0434*** (0.0160)	-0.0302** (0.0148)	-0.0136 (0.0125)	-0.0017 (0.0199)	0.0038 (0.0168)	NA	NA

PANEL E: Pharmaceutical Sector Firms Results								
Firm Names	13th Lok Sabha Elections	13th Lok Sabha Elections	14th Lok Sabha Elections	14th Lok Sabha Elections	15th Lok Sabha Elections	15th Lok Sabha Elections	16th Lok Sabha Elections	16th Lok Sabha Elections
	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2
Dr. Reddy'S Laboratories Ltd.	Not Significant	Not Significant	Medium	Not Significant	High	Medium	High	High
	-0.0017 (0.0156)	0.0039 (0.0132)	0.0270** (0.0115)	0.0147 (0.00974)	0.0366*** (0.0127)	0.0252** (0.0108)	-2.011*** (0.665)	-1.496*** (0.567)

PANEL F: Infrastructure Sector Firms Results								
Firm Names	13th Lok Sabha Elections	13th Lok Sabha Elections	14th Lok Sabha Elections	14th Lok Sabha Elections	15th Lok Sabha Elections	15th Lok Sabha Elections	16th Lok Sabha Elections	16 th Lok Sabha Elections
	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2	Dummy 1	Dummy 2
Ansal Properties & Infrastructure Ltd.	Not Significant	Not Significant	NA	NA	High	High	High	High
	-0.0111 (0.0266)	0.0147 (0.0226)	NA	NA	0.0532*** (0.0196)	0.0532*** (0.0165)	0.0534*** (0.0153)	0.0383*** (0.0131)
DLF Ltd.	NA	NA	NA	NA	Not Significant	Not Significant	High	High
	NA	NA	NA	NA	0.0195 (0.0175)	0.0174 (0.0148)	0.0331*** (0.0118)	0.0384*** (0.00992)

Era Infra Engg. Ltd.	NA	NA	NA	NA	Not Significant	High	High	High
	NA	NA	NA	NA	0.0165 (0.0143)	0.0401*** (0.0119)	0.0400*** (0.0146)	0.0437*** (0.0123)
Gammon India Ltd.	Not Significant	Not Significant	Not Significant	Not Significant	High	High	High	High
	0.000489 (0.0165)	-0.00240 (0.0140)	-0.0203 (0.0137)	-0.0140 (0.0115)	0.0934*** (0.0181)	0.0767*** (0.0153)	5.714*** (1.505)	6.083*** (1.259)
G M R Infrastructure Ltd.	NA	NA	NA	NA	High	Medium	Medium	High
	NA	NA	NA	NA	0.0508*** (0.0161)	0.0301** (0.0137)	0.0276** (0.0117)	0.0260*** (0.00991)
G V K Power & Infrastructure Ltd.	NA	NA	NA	NA	Low	Low	High	High
	NA	NA	NA	NA	0.0321* (0.0192)	0.0291* (0.0162)	0.0395*** (0.0150)	0.0364*** (0.0127)

Hindustan Construction Co. Ltd.	Not Significant	Not Significant	Not Significant	Not Significant	High	High	High	High
	-0.0223 (0.0177)	-0.0152 (0.0150)	-0.0050 (0.0144)	0.0034 (0.0121)	0.0583*** (0.0189)	0.0547*** (0.0159)	5.919*** (1.596)	4.868*** (1.357)
I V R C L Infrastructures & Projects Ltd.	NA	NA	Not Significant	Not Significant	Medium	High	High	High
	NA	NA	-0.0137 (0.0174)	-0.0001 (0.0147)	0.0429** (0.0206)	0.0473*** (0.0173)	9.177*** (1.511)	6.313*** (1.312)
	NA	NA	NA	NA	Not Significant	Not Significant	High	High
Lanco Infratech ltd	NA	NA	NA	NA	0.0294 (0.0205)	0.0271 (0.0173)	8.284*** (1.615)	8.491*** (1.341)
Omaxe Ltd.	NA	NA	NA	NA	High	High	Medium	High
	NA	NA	NA	NA	0.0700*** (0.0134)	0.0613*** (0.0113)	0.0139** (0.00616)	0.0149*** (0.00520)

Peninsula land ltd	Not Significant	Not Significant	Not Significant	Not Significant	Not Significant	Not Significant	High	High
	-0.0125 (0.0249)	0.0025 (0.0211)	-0.0121 (0.0213)	-0.0121 (0.0179)	0.0234 (0.0194)	0.0212 (0.0164)	0.0446*** (0.0141)	0.0383*** (0.0119)
Unitech ltd	Not Significant	Not Significant	Not Significant	Not Significant	Not Significant	Not Significant	High	High
	0.0099 (0.0141)	0.0107 (0.0120)	0.0041 (0.0162)	0.0076 (0.0136)	0.0269 (0.0247)	0.0294 (0.0209)	0.0634*** (0.0150)	0.0602*** (0.0126)

The multivariate regression was run separately for each of the sectors (appendix I). We also provide a consolidated table (Table III) presenting the significant results for the sake of simplicity and understanding.

When we look at the firm level coefficients for the event dummies, we find that the impact on the firms ranges from high to low level of significance for the firms. Table III presents the coefficients and their significance level industry wise. We can see that the results have become significant for more number of firms in the recent years over the earlier years. We have more coefficients coming significant for our 4th event which is the 16th Lok Sabha election results. Further, we can see that infrastructure and metal stocks' coefficient for the dummy variables is coming highly significant for the fourth event than for the first, second and third event. A possible reason could be that markets over time have learned to process the information regarding political ties or that due to more information about political parties and businesses being available, investors' perception about the stocks has been formed and thus gets reflected in the returns. Further, the historical win of Modi government during the 16th Lok Sabha election and the BJP party coming into power gave a boost to the economy and gained the trust of many investors. Another possible reason could be that – it was a possible one off aberration – wherein markets were completely surprised by the direction of the results and thus resulting in (visible) relationships.

The coalition which formed the government after the 13th Lok Sabha elections was led by Bhartiya Janata Party (see Table I). For the 13th Lok Sabha election result event, we find that Bank of Baroda was impacted positively (see Table III, Panel C) but at a medium level (5 percent level of significance). The event had a medium (5percent level of significance) negative impact on Britannia Industries Ltd. (see Table III, Panel) while had a high (1 percent level of significance) positive impact on Gujarat Mineral Development Corporation Ltd. (see Table III Panel C). Hence for these companies, we reject the null hypothesis and accept that the event has an impact on the stock returns.

In the next elections, the 14th Lok Sabha elections, the coalition which came to power was led by Indian National Congress. This was a regime change. For this event, we find that Tata Steel is impacted by the event at a highly significant level. Gujarat Mineral Development Corporation Ltd. and State Bank of India is impacted negatively at a medium level while Dr. Reddy's Laboratories Ltd show medium negative impact. HDFC bank and Glaxosmithkline Pharmaceuticals Ltd. are impacted by the event but at a low level.

Our third event is the 15th Lok Sabha election results. The winning coalition was again led by Indian National Congress. During this period, we find a lot more companies' returns being affected by the event. The companies which show a low degree of positive impact of the event include GVK Power & Infrastructure Ltd., Bank of Baroda. The companies showing medium level of positive impact include State Bank of India. The companies which show high degree of positive impact include: Infrastructure firms – Ansal Properties & Infrastructure Ltd., Gammon India Ltd., Hindustan Construction Co. Ltd., Omaxe Ltd., Media firms – H T Media Ltd., Jagran Prakashan Ltd.

For this event, Dr. Reddy's Laboratories Ltd. stock shows a High degree of positive impact for the first dummy (-1, +3 days) and medium degree of impact for our second dummy (-1,+5 days). Similar results are also seen for Bank of India and GMR Infrastructure Ltd. On the other way, IVRCL Infrastructure and Projects Ltd. show a medium positive impact in the first dummy variable and a high positive impact for the second dummy variable.

Our results help in identifying the stocks which have an impact because of the election results news entering the market. The results can be analysed keeping in mind the connectedness that the market perceives the firm might be having with the coalition formed after the announcement of the election results. With the increasing disclosure norms in the Indian political environment, there is access to information regarding donations/contributions to the political parties which might have led to the impact on the stock market returns of few firms. Also, political leaders being on boards and getting

involved in the business activities might be another reason for this perception. The increasing role of media (Appendix II) also might have led to the perception of political connectedness of the firms. The coverage about businesses and their closeness to the regional/coalition political party helps in explaining the signs we are getting for our coefficients. For a few firms, we are able to capture the change in signs because of the regime change.

Significant results of the multivariate regression for the firms for the event 13th, 14th, 15th and 16th Lok Sabha election result. **Dummy1** is the dummy for the event window (-1,+3), **dummy2** is the dummy for the event window (-1,+5). The degree of impact is measured using the significance of the t-statistic for the coefficient from the regression results mentioned in parenthesis. *** $p < 0.01$ (high), ** $p < 0.05$ (medium), * $p < 0.1$ (low). The signs of the coefficients are mentioned in parenthesis.

If we look at the results, we find that Gujarat Mineral Development Corporation has a positive impact during the 13th Lok Sabha election results. The possible explanation for the same could be the presence of same party in power at the centre and at the state level. However, it loses its significance in 15th Lok Sabha results. When we carefully look at the 15th Lok Sabha election results, we find that some media companies such as NDTV HT media showing significance of the coefficients for the dummy variables. Perhaps the time taken to form a coalition and to filter information about allotment of key ministries is captured by our dummy variables.

The results have the following implications for investors and portfolio managers:

- a. During the period when the election results are declared, the investors can choose their investment strategy based on the degree of impact that the firms have with the election results. After knowing the coalition or party in power and the information surrounding the company, they can take advantage of the returns during the event period.
- b. Portfolio Managers can look for the firms which are displaying high/low returns during our event windows (-1,+3 days) and (-1,+5 days) and decide the portfolio changes according to the results of the elections. Similarly, they can take a buy-sell decision during this period based on the party coming to power. Most likely, the regime change could also be factored in looking at the results.

CONCLUSION AND AREAS OF FURTHER WORK

Indian stock markets are affected by different events and these can be captured by undertaking an analysis of the abnormal returns during the events. In our study, we use a national event, election results announcement, and try to create a metric for different firms. We use an application of multivariate regression model with dummy variables for the event date (Schipper and Thompson, 1983, 1985) and also the cumulative abnormal returns are calculated using the market model. At many instances, the results are found to be significant for both the multivariate regression and cumulative abnormal returns.

We find that over the years, the stock market have possibly become more efficient at information processing about the election results. On the basis of the market reaction to the event, we put the firms in three categories of political impact for the three election results – low, medium and high. We summarize the results of multivariate regression, cumulative abnormal return (CAR) as follows:

Table IV. Summary of the Results

	Low Degree of Connectedness	Medium Degree of Connectedness	High Degree of Connectedness
13th Lok Sabha Elections results	-	Positive- Bank of Baroda Negative- Britannia Industries Ltd.	Positive – Gujarat Mineral Development Corporation Ltd.
14th Lok Sabha Elections results	-	-	Positive – Tata Steel Ltd.
15th Lok Sabha Election results	Positive – GVK Power & Infrastructure Ltd., Bank of Baroda	Positive- GMR Infrastructure Ltd., Bank of India, Dr. Reddy's Laboratories Ltd., State Bank of India	Positive - Ansal Properties & Infrastructure Ltd., Balaji Telefilms Ltd., Gammon India Ltd., Hindustan Construction Co. Ltd., I V R C L Infrastructures & Projects Ltd., Era Infra Engg. Ltd., Omaxe Ltd., Balaji Telefilms

			Ltd., H T Media Ltd., Jagran Prakashan Ltd., New Delhi Television Ltd.
16th Lok Sabha Election results		Positive- Federal Bank Ltd.	Positive- Ansal Properties & Infrastructure Ltd., DLF Ltd., Era Infra Engg. Ltd., Gammon India Ltd., GMR Infrastructure Ltd., Hindustan Construction Co. Ltd., I V R C L Infrastructures & Projects Ltd., Omaxe Ltd., Canara Bank, Punjab National Bank, State Bank of India, ITC Ltd., Jindal Saw Ltd., National Aluminum Co Ltd., Hathway Cable Datacom Ltd., Peninsula Land Ltd., Lanco Infratech Ltd., Bhushan Steel Ltd

In this study, we present the analysis for a few selected firms and sectors listed on National Stock Exchange. There is scope to do this analysis for a larger number of firms. There is a possibility to look at the coefficients at industry level also. Though there is a challenge in defining political connectedness, the board composition of these firms could be looked at to ascertain connectedness. However, in a country context such as India with large lack of transparency in political-business dealings, we doubt if that will ever make things more clear. We have looked at one metric of knowing this, which is stock returns behaviour, during the political event – there could possibly other barometers to capture these – in terms of the private rent seeking opportunities accruing to the business owners and promoters (and not to necessarily to all the shareholders of a firm). For example, new project approvals accruing to the business group could be diverted to potential private parties owned by one key owner. Similarly, there could be other scenarios – wherein one stakeholder takes much more benefits of these political connectedness – which are invisible in nature – and not available to most of the other stakeholders as well.

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Notes

1 <http://business.rediff.com/slide-show/2009/may/18/slide-show-1-sensex-rockets-on-upa-victory.htm>

2 <http://m.indianexpress.com/news/the-businesspolitics-nexus/971862/>

3 See Getz (1993). Getz(1993) categorizes firm political activities into seven broad categories which include these commonly studies activities: lobbying, reporting, research results, reporting survey results, testifying at government hearings, legal actions, personal service and constituency building.

4 See Thakur (1989), *Business and Politics in India: A Historical Perspective*, Economic and Political Weekly, Vol. 24, No. 26 (July 1, 1989: 1437-1439).

5 Gowda et. al further add the reasoning for businesses being secretive. They add, ‘Maintaining confidentiality of donations helps avoid reprisals by political parties that might want to penalize the donor for favoring their opponents; this is generally regarded as more important than any tax benefit.’

6 Jaypee group and its proximity to Bahujan Samaj Party, Jindals and Congress, Lanco Infratech and Congress, NDTV and Congress are a few which have had a mention in media.

7 See ‘*Analysis of income tax returns filed and donations received by political parties :A Report by National Election Watch & Association for Democratic Reforms*’ ,
(<http://adrindia.org/sites/default/files/Report%20donations.pdf>)

8 While political affiliations with the ruling party do not necessarily imply favouritism in all public governance contexts, it is possible that this may happen. And this may happen more often than not in a country context such as India – given the nature of public governance structures (for example, please see, Khanna, Tarun, and Krishna G. Palepu. ‘*Why Focused Strategies May Be Wrong for Emerging Markets.*’ Harvard Business Review 75, no. 4 (July–August 1997: 41–51).

9 Though we present the results for the $(-1,+3)$ and $(-1, +5)$ event windows, we ran our model for $(-1,+1)$, $(-3, +3)$, $(-5, +5)$ event windows also. The reason for choosing the $(-1,+3)$ and $(-1, +5)$ was to factor in the time taken to form the coalition after the election results date and the coalition politics results in more time taken for the equations to finally firm up.

APPENDIX

I. List of Firms

Infrastructure	Banking	Media	FMCG	Metal	Pharmaceutical
Ansal Properties & Infrastructure Ltd.	Axis Bank Ltd.	Adlabs Films Ltd.	Britannia Industries Ltd.	Bhushan Steel Ltd.	Aurobindo Pharma Ltd.
D L F Ltd.	Bank Of Baroda	Balaji Telefilms Ltd.	Colgate-Palmolive (India) Ltd.	Gujarat Mineral Devp. Corpn. Ltd.	Biocon Ltd.
Era Infra Engg. Ltd.	Bank Of India	Deccan Chronicle Holdings Ltd.	Dabur India Ltd.	Gujarat N R E Coke Ltd.	Cadila Healthcare Ltd.
G M R Infrastructure Ltd.	Canara Bank	Den Networks Ltd.	Emami Ltd.	Hindalco Industries Ltd.	Cipla Ltd.
G V K Power & Infrastructure Ltd.	Federal Bank Ltd.	Dish T V India Ltd.	Glaxosmithkline Consumer Healthcare Ltd.	J S W Steel Ltd.	Divi'S Laboratories Ltd.
Gammon India	H D F C Bank	Eros	Godrej	Jindal Saw	Dr.

Ltd.	Ltd.	International Media Ltd.	Consumer Products Ltd.	Ltd.	Reddy'S Laborator ies Ltd.
Hindustan Construction Co. Ltd.	I C I C I Bank Ltd.	Hathway Cable & Datacom Ltd.	Godrej Industries Ltd.	Jindal Steel & Power Ltd.	Glaxosmi thkline Pharmace uticals Ltd.
Housing Development & Infrastructure Ltd.	Indusind Bank Ltd.	H T Media Ltd.	Hindustan Unilever Ltd.	Maharashtra Seamless Ltd.	Glenmark Pharmace uticals Ltd.
I V R C L Infrastructures & Projects Ltd.	Kotak Mahindra Bank Ltd.	Inox Leisure Ltd.	I T C Ltd.	Monnet Ispat & Energy Ltd.	Lupin Ltd.
Jaiprakash Associates Ltd.	Punjab National Bank	Jagran Prakashan Ltd.	Jubilant Foodworks Ltd.	National Aluminium Co. Ltd.	Piramal Healthcar e Ltd.
Lanco Infratech Ltd.	State Bank Of India	Network Returns	Marico Ltd.	Sesa Goa Ltd.	Ranbaxy Laborator ies Ltd.
Nagarjuna Construction Co. Ltd.	Yes Bank Ltd.	New Delhi Television Ltd.	Mcleod Russel India Ltd.	Steel Authority Of India Ltd.	Sun Pharmace utical Inds. Ltd.

Omaxe Ltd.		P V R Ltd.	Procter & Gamble Hygiene & Health Care Ltd.	Sterlite Industries (India) Ltd.	var1
Peninsula Land Ltd.		Pyramid Saimira Theatre Ltd.	Tata Global Beverages Ltd.	Tata Sponge Iron Ltd.	
Unitech Ltd.		Sun T V Network Ltd.	United Spirits Ltd.	Tata Steel Ltd.	
		T V Today Network Ltd.		var1	
				Vedanta Ltd.	
				Welspun Corp Ltd.	
15	12	16	15	18	13